MCSA Windows Server

Lab Project Documentation

ITI Intensive Program — Intake 46

System Administration Track

PROJECT TEAM

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1. Active Directory Domain Services (AD DS)

The project is built on a Windows Server 2022 Active Directory forest named ITI.LOCAL. DC1 serves as the Primary Domain Controller (PDC), hosting AD DS, DNS, and DHCP roles. Two child domains were created under the main forest: Alex.ITI.LOCAL and Ismaliya.ITI.LOCAL.

1.1 DC1 — Primary Domain Controller Dashboard

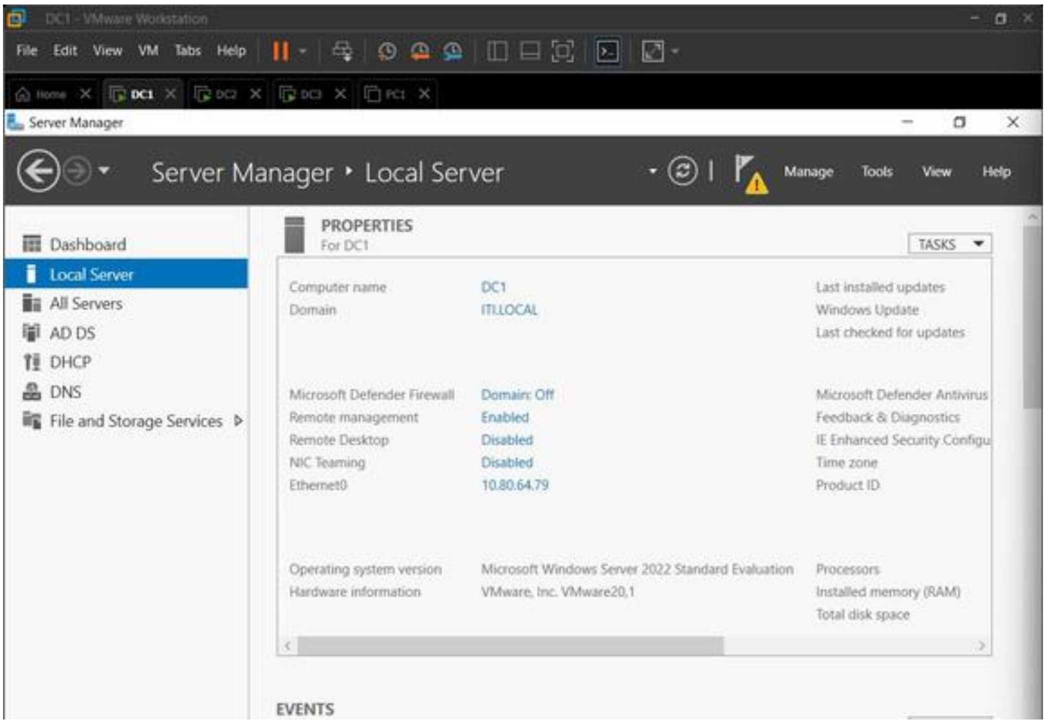


Figure: DC1 Local Server properties — Windows Server 2022, domain ITI.LOCAL, IP 10.80.64.79

1.2 The Three Domain Controllers (DC1, DC2, DC3/RODC)

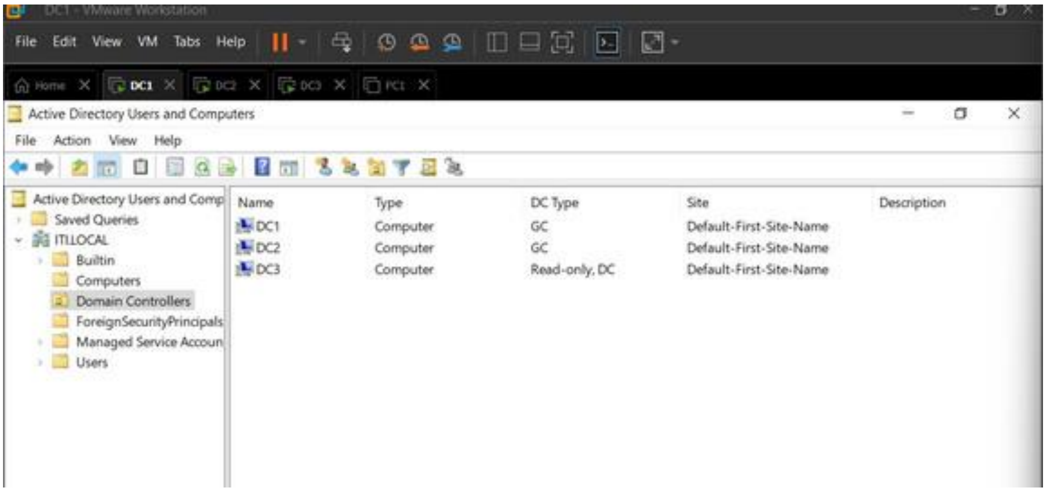


Figure: Domain Controllers OU showing DC1, DC2 (both GC), and DC3 as Read-Only DC (RODC)

1.3 RODC — Read-Only Domain Controller

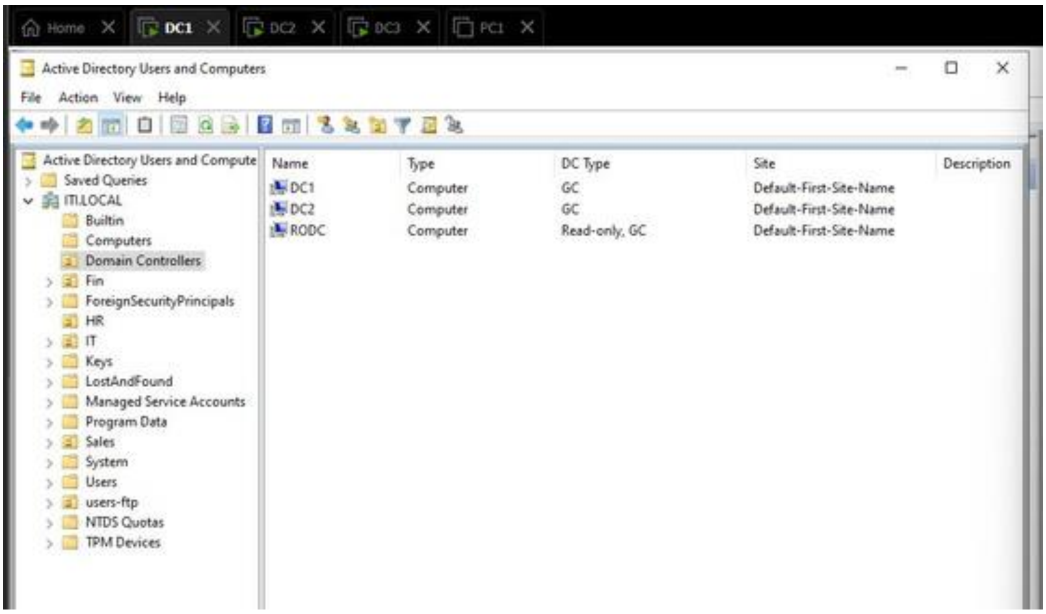


Figure: RODC is configured as a Read-Only, GC domain controller under ITI.LOCAL — used for branch office security

2. Child Domain Configuration

Two child domains were joined to the ITI.LOCAL forest. Alex.ITI.LOCAL represents the Alexandria branch, and Ismaliya.ITI.LOCAL represents the Ismailia branch. Each child domain has its own Domain Controller registered under the parent forest.

2.1 Alex and Ismaliya Domains Joined to Main Forest

Figure: Active Directory Domains and Trusts — Alex.ITI.LOCAL and Ismaliya.ITI.LOCAL registered under ITI.LOCAL

2.2 Ismaliya Domain Controller — Server Manager

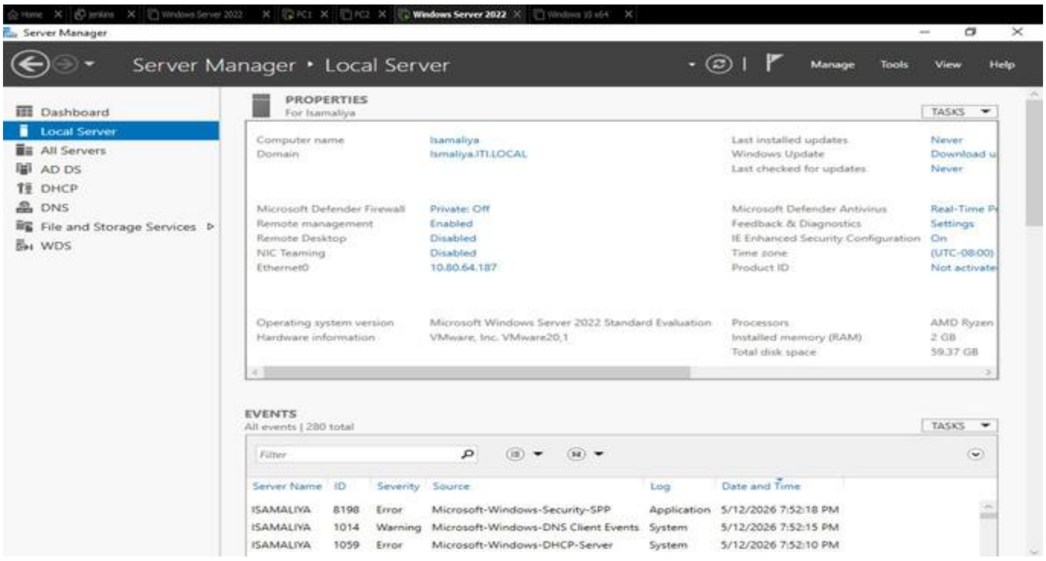


Figure: Isamaliya server joined to Ismaliya.ITI.LOCAL child domain with AD DS, DHCP, DNS, and WDS roles

3. Organizational Units & User Accounts

The AD structure includes dedicated Organizational Units (OUs) for each department: HR, IT, Finance (Fin), and Sales. Users are created inside their respective OUs. A dedicated OU named users-ftp was also created for FTP service accounts. A bulk CSV script was used to create users across multiple OUs automatically.

3.1 HR Department Users

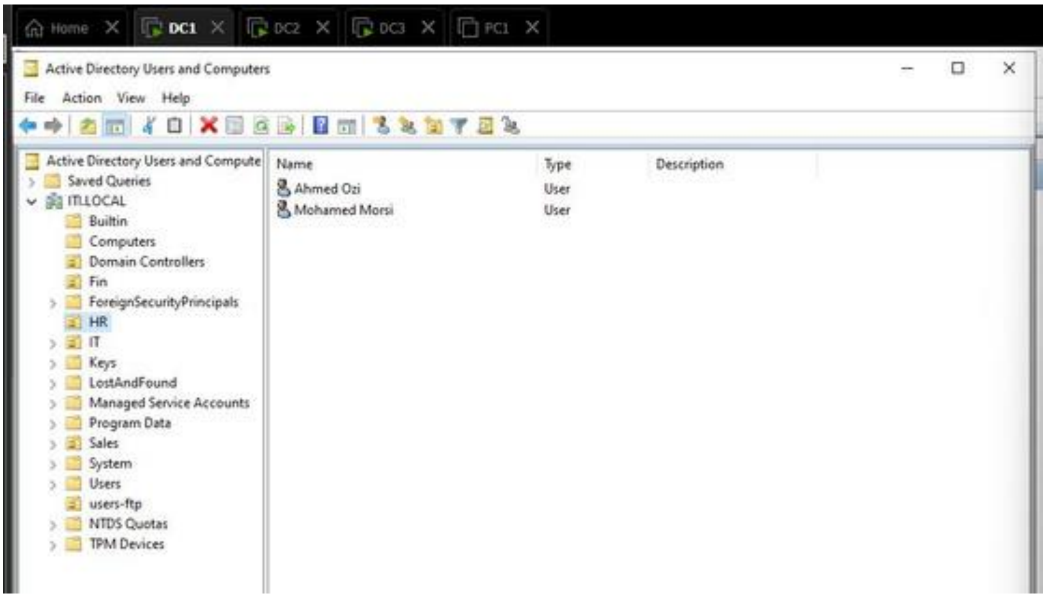


Figure: HR OU containing users Ahmed Ozi and Mohamed Morsi

3.2 IT Department Users

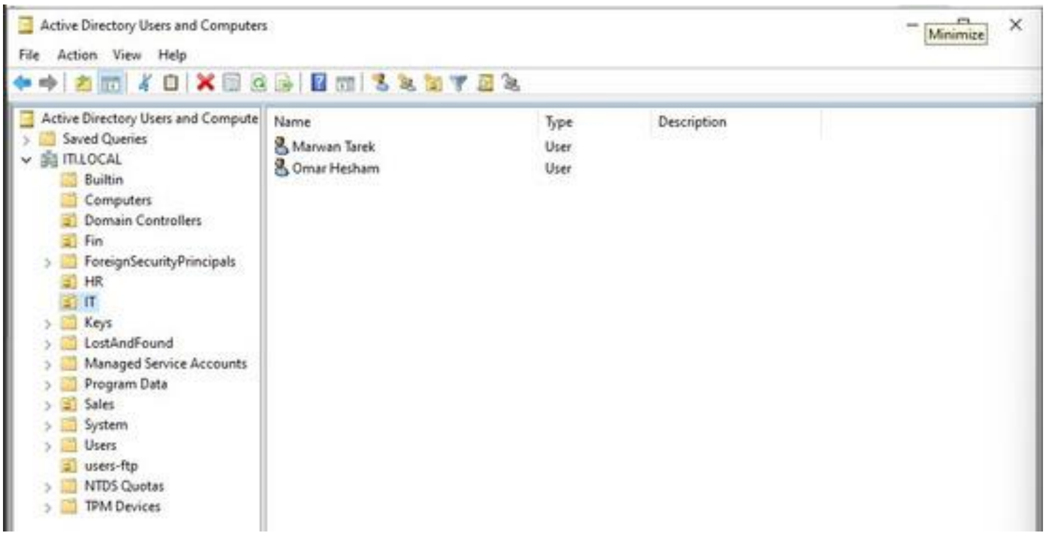


Figure: IT OU containing users Marwan Tarek and Omar Hesham

3.3 Finance Department User

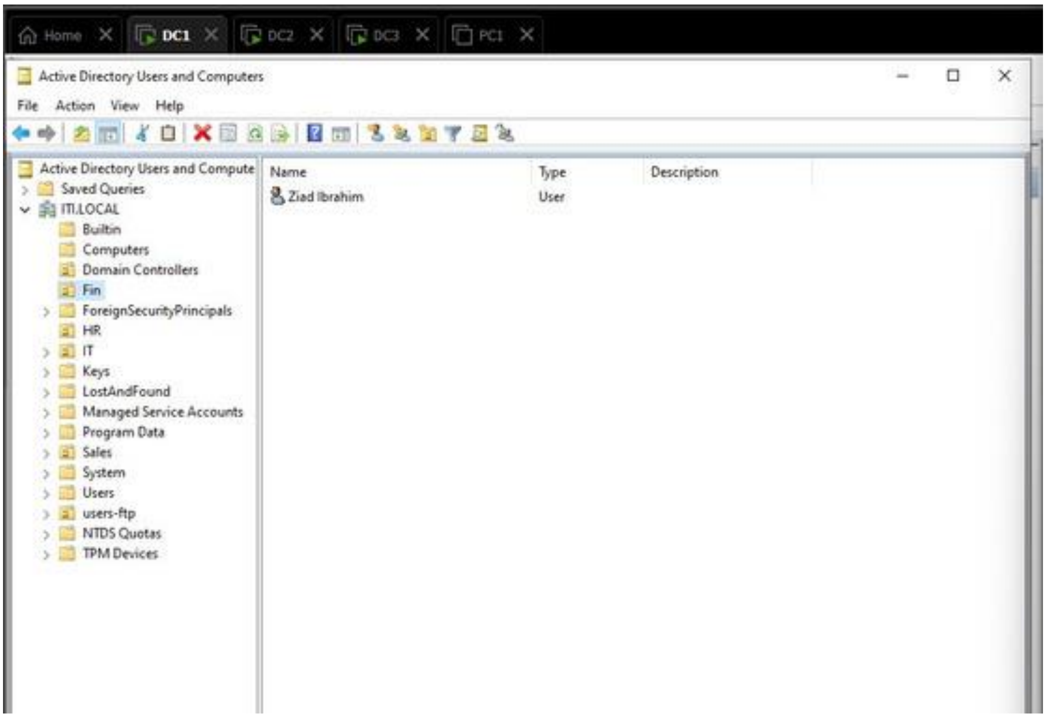


Figure: Fin OU containing user Ziad Ibrahim

3.4 FTP Dedicated User

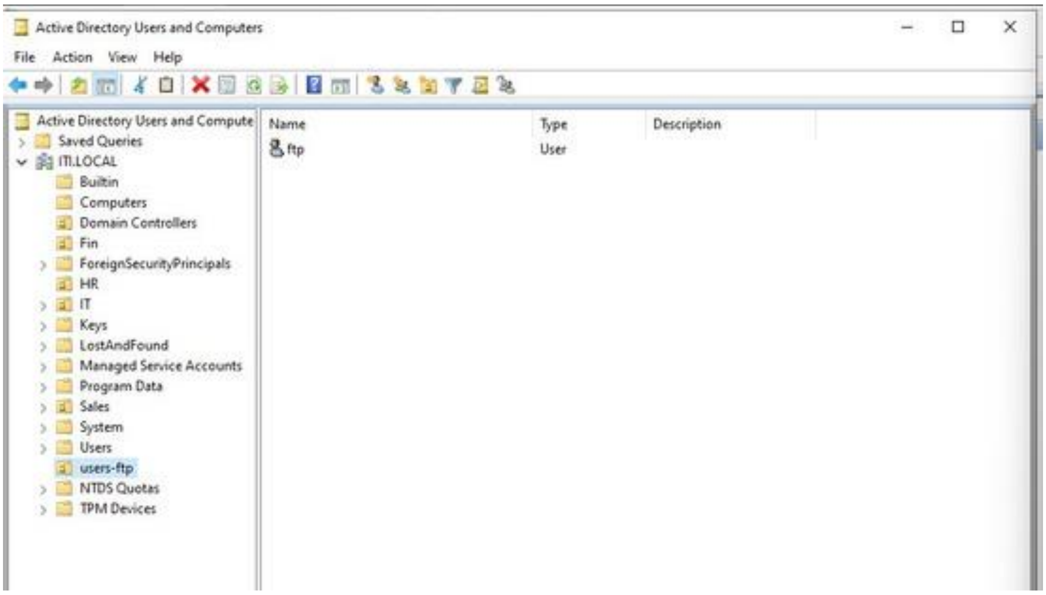


Figure: users-ftp OU with a dedicated "ftp" service account for FTP authentication

3.5 Bulk User Creation via CSV Script

```
FirstName,LastName,Username,Password,OU
Ahmed,Mansour,ahmed.mansour,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Sara,Hassan,sara.hassan,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Nour,Ibrahim,nour.ibrahim,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Mona,Khalil,mona.khalil,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Layla,Farouk,layla.farouk,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Hana,Samir,hana.samir,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Rania,Mostafa,rania.mostafa,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Dina,Adel,dina.adel,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Yasmin,Tarek,yasmin.tarek,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Omar,Recruiter,omar.recruiter,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Khaled,Nasser,khaled.nasser,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Tariq,Badawi,tariq.badawi,P@$$w0rd,OU=HR,DC=ITI,DC=LOCAL
Karim,Sales,karim.sales,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Mohamed,Gamal,mohamed.gamal,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Youssef,Fathi,youssef.fathi,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Amr,Zaki,amr.zaki,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Hassan,Lotfy,hassan.lotfy,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Sherif,Ragab,sherif.ragab,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Adel,Wahba,adel.wahba,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Mahmoud,Saber,mahmoud.saber,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Ramadan,Helmy,ramadan.helmy,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Walid,Shafik,walid.shafik,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Bassem,Gouda,bassem.gouda,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Nabil,Tawfik,nabil.tawfik,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Tarek,Hamdy,tarek.hamdy,P@$$w0rd,OU=Sales,DC=ITI,DC=LOCAL
Ali,Sysadmin,ali.sysadmin,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Ahmed,Devops,ahmed.devops,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Mariam,Network,mariam.network,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Samir,Security,samir.security,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Hossam,Helpdesk,hossam.helpdesk,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Rami,Database,rami.database,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Nada,Cloud,nada.cloud,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Fady,Developer,fady.developer,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
Bishoy,Support,bishoy.support,P@$$w0rd,OU=IT,DC=ITI,DC=LOCAL
```

Figure: CSV file used to bulk-create users across HR, Sales, and IT OUs with predefined passwords and OU paths

3.6 Computers Registered in DC1

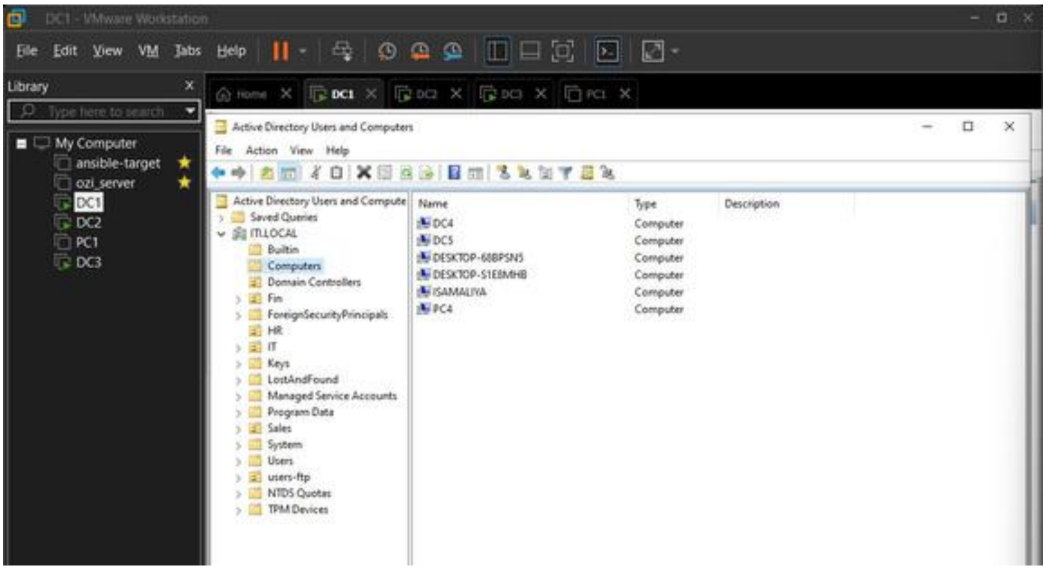


Figure: Computers OU showing all joined machines: DC4, DC5, two desktops, ISAMALIYA, and PC4

4. DHCP Server Configuration

DHCP was configured on DC1 for the ITI.LOCAL domain and separately on the Ismaliya server for the 192.168.70.0/24 subnet. Scope options include IP range, lease duration, and PXE boot option (060 PXEClient) for WDS deployment.

4.1 DHCP Scope on DC1 (10.0.0.0 Network)

Figure: DC1 DHCP IPv4 scope [10.0.0.0] — Active status confirming the scope is live

4.2 DHCP Address Pool (Ismaliya Server)

Figure: Address Pool range: 192.168.70.150 – 192.168.70.250 for client subnet distribution

4.3 DHCP Address Lease

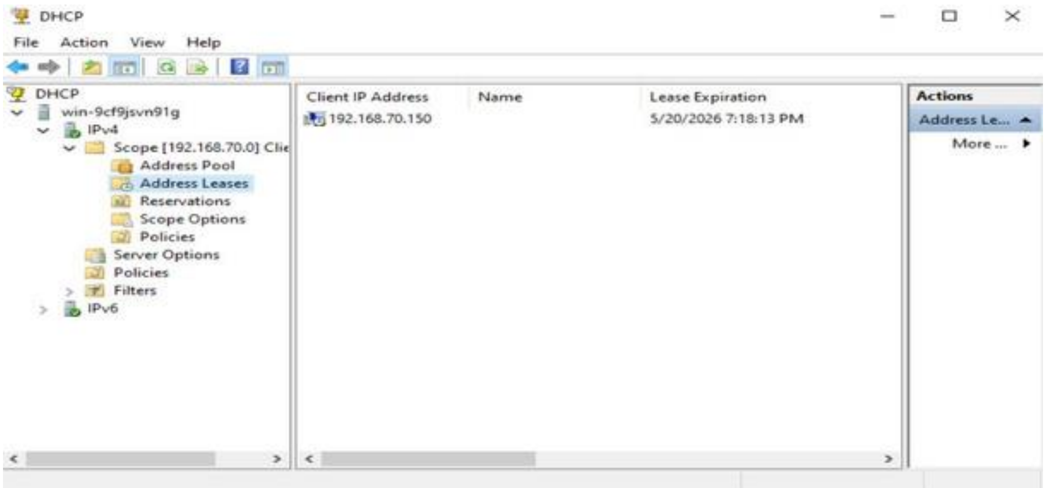


Figure: Active lease showing 192.168.70.150 assigned to a client — lease expiry 5/20/2026

4.4 DHCP Server Options — PXE Boot

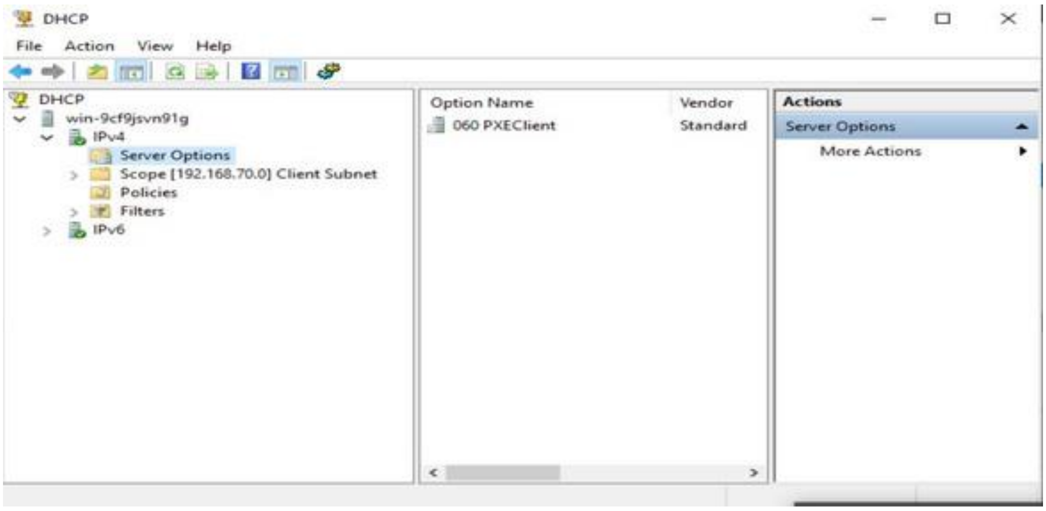


Figure: Server Options showing "060 PXEClient" configured to support WDS network boot

5. Joining Client Machines to Domains

Client machines (PCs) were joined to their respective domains. PC4 was joined to the Alex.ITI.LOCAL child domain, and PC5 was joined to the Ismaliya.ITI.LOCAL child domain. PC1 was also configured with a static DNS pointing to DC1.

5.1 PC1 Network Configuration

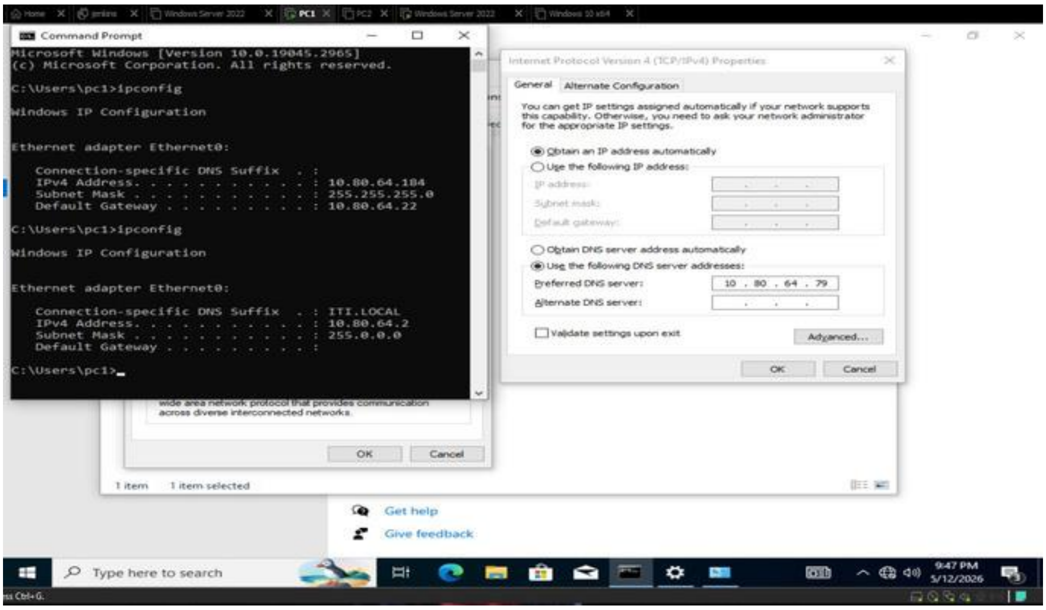


Figure: PC1 set to automatic IP from DHCP, with Preferred DNS manually set to 10.80.64.79 (DC1)

5.2 PC4 Joined to Alex.ITI.LOCAL

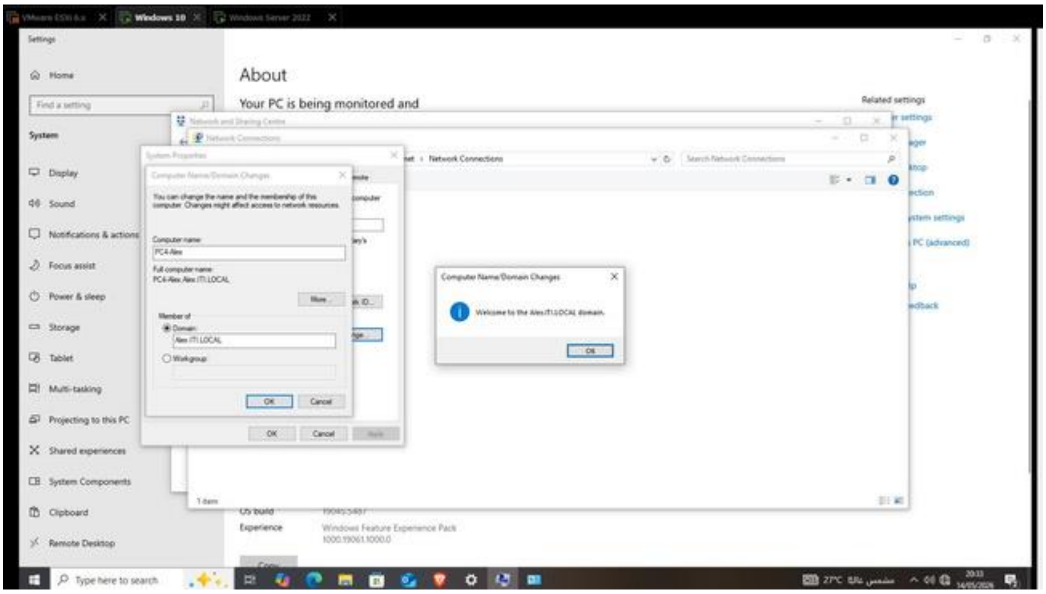
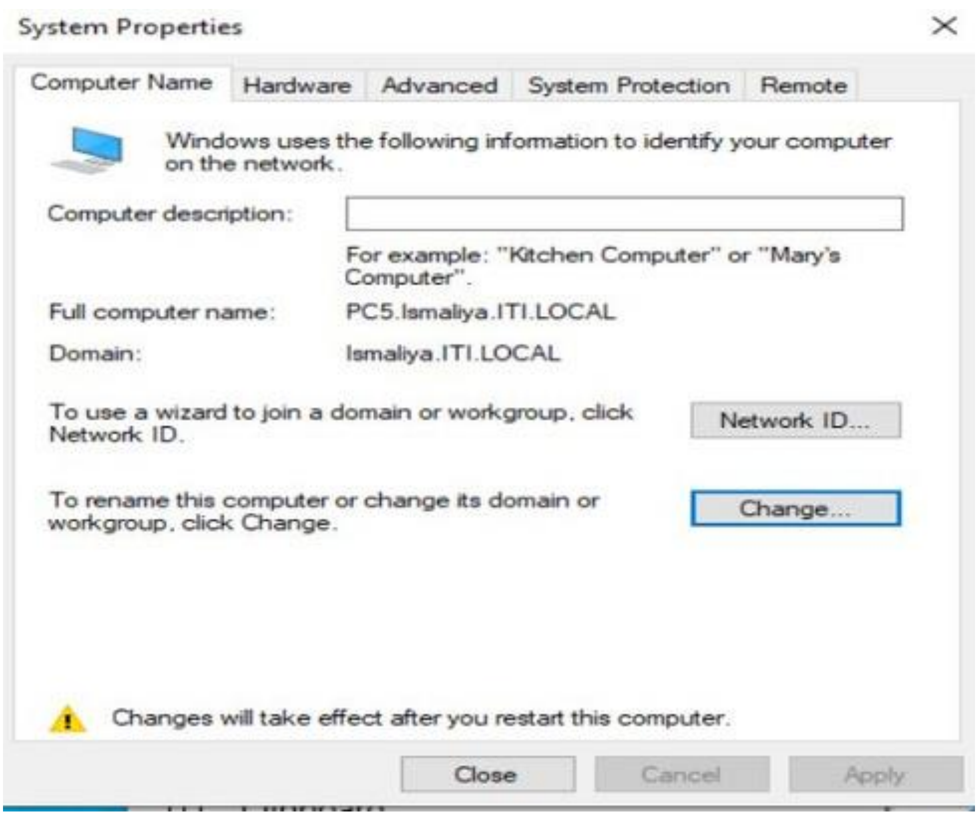


Figure: PC4-Alex successfully joined Alex.ITI.LOCAL — "Welcome to the Alex.ITI.LOCAL domain" confirmation

5.3 PC5 Joined to Ismaliya.ITI.LOCAL



6. Remote Desktop Protocol (RDP)

RDP was tested between domain machines. A connection was established from a client PC to DC4 (Alex.ITI.LOCAL) using the domain credentials. The certificate warning appears because no CA-signed certificate was installed — this is expected in a lab environment.

6.1 RDP Connection Attempt to DC4

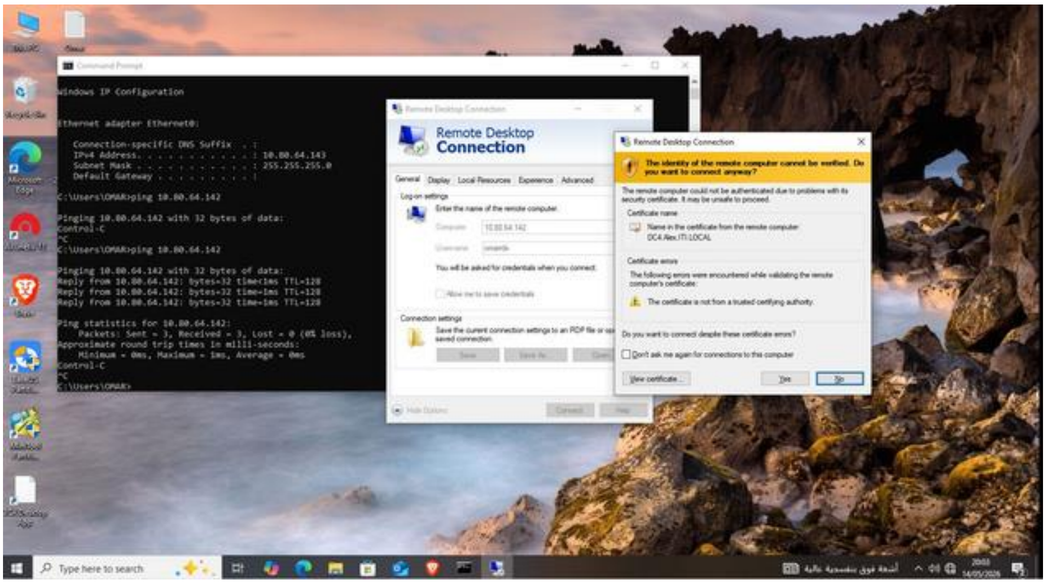


Figure: RDP session initiated to 10.80.64.142 — certificate warning shown for DC4.Alex.ITI.LOCAL (expected in lab)

6.2 Successful RDP Session

Figure: Active RDP session to 10.80.64.142 showing Windows 10 desktop — remote management working

7. Group Policy Objects (GPO)

Group Policies were applied at the domain level to enforce security controls across all domain-joined machines. Three main policies were configured: disabling Control Panel access, blocking all USB/removable storage devices, and forcing a corporate wallpaper on all user desktops.

7.1 Disabling Control Panel via GPO

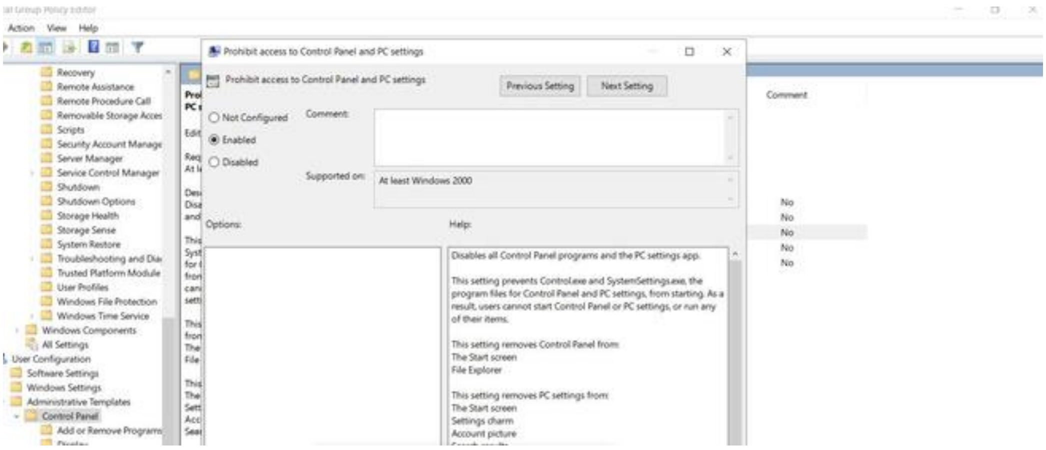


Figure: GPO setting "Prohibit access to Control Panel and PC settings" — set to Enabled

7.2 Control Panel Blocked — Result on Client

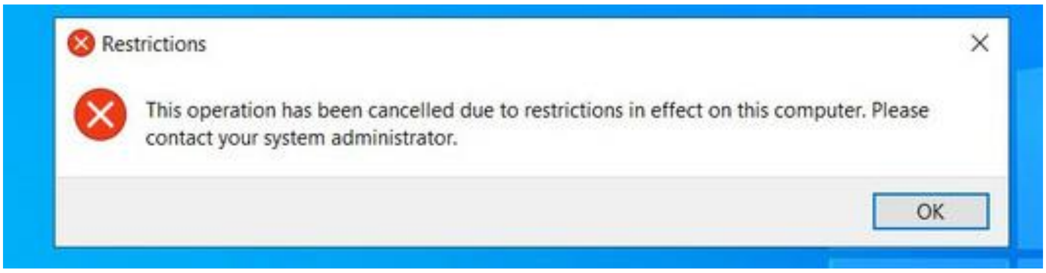


Figure: Restrictions pop-up confirming Control Panel is blocked — "operation cancelled due to restrictions"

7.3 Disabling USB / Removable Storage

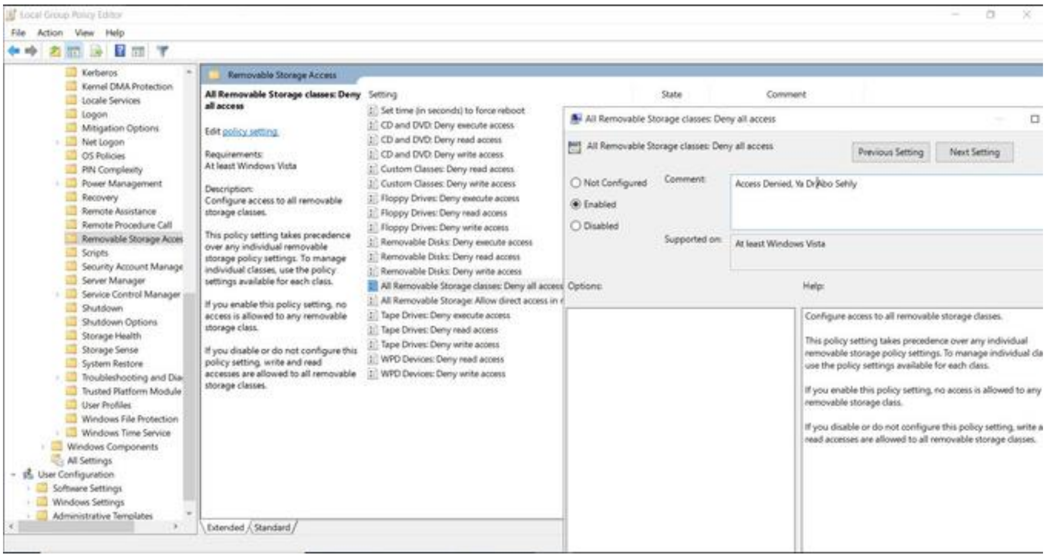


Figure: "All Removable Storage classes: Deny all access" enabled — blocks USB, floppy, and removable disks

7.4 USB Access Denied — Result on Client

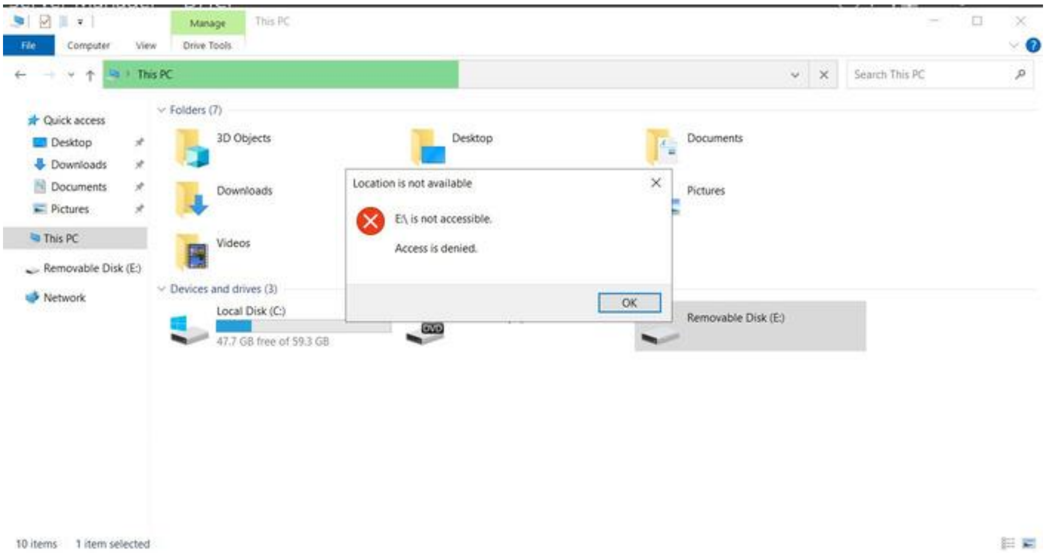


Figure: "E:\ is not accessible — Access is denied" error confirming USB is blocked by GPO policy

7.5 Forcing Corporate Wallpaper

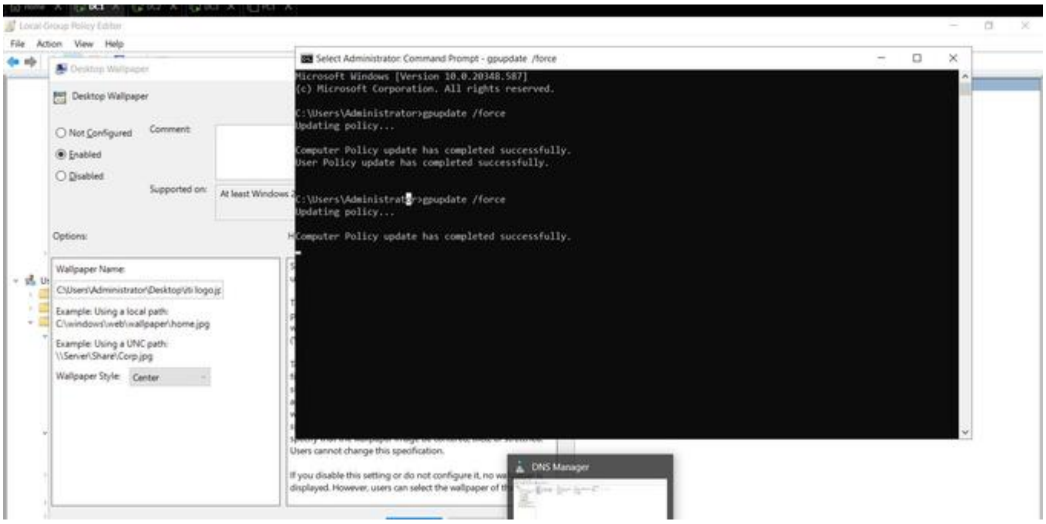


Figure: Desktop Wallpaper GPO set to Enabled with path to ITI logo — gpupdate /force applied successfully

7.6 Wallpaper Applied on Client

Figure: ITI Alexandria Branch logo displayed as the forced wallpaper on the domain-joined desktop

8. IIS Web Server & FTP Site

IIS (Internet Information Services) was installed on DC5 to host two internal websites (web1.com and web2.com) and one FTP site. Websites are mapped to C:\mywebsites\web1 and web2 directories. The FTP site is bound to www.ftp.com and mapped to C:\mywebsites\ftp.

8.1 IIS Sites Overview

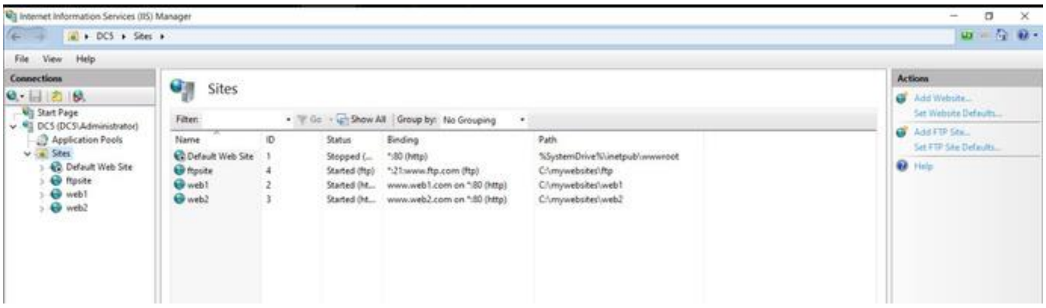


Figure: IIS Manager showing 4 sites: Default Web Site (stopped), ftpsite, web1 (www.web1.com:80), web2 (www.web2.com:80)

8.2 IIS Initial Setup

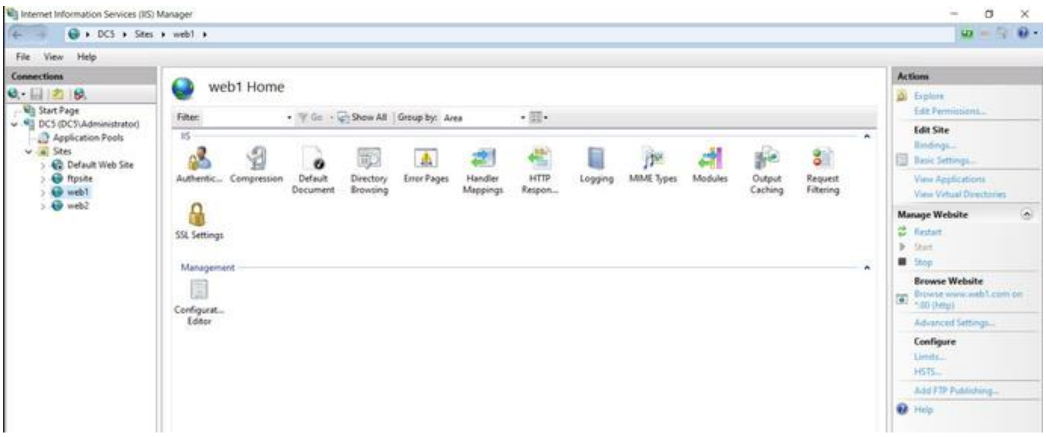


Figure: Early IIS setup showing the first site configuration before adding all virtual hosts

8.3 web1 Site Home

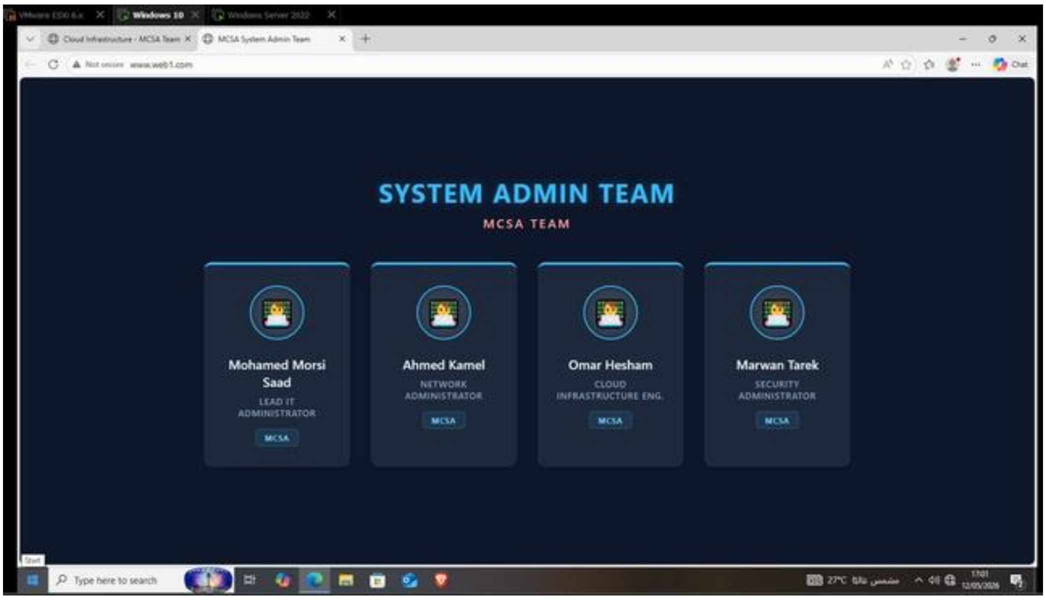


Figure: www.web1.com — System Admin Team page served from IIS, showing all 4 MCSA team members

8.4 web2 Site Home

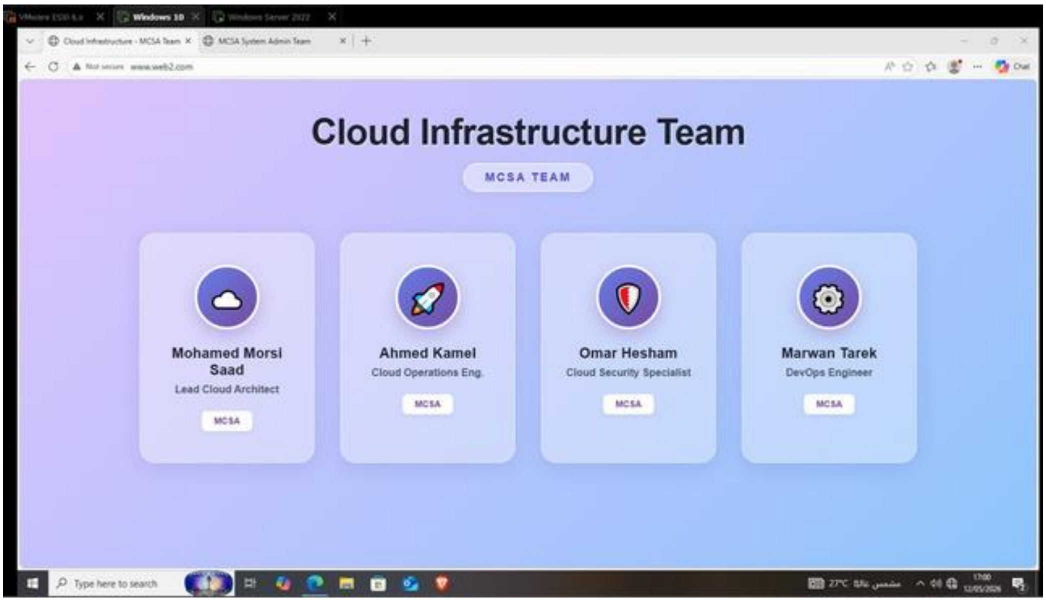


Figure: www.web2.com — Cloud Infrastructure Team page with team roles displayed correctly

8.5 FTP Folder Structure

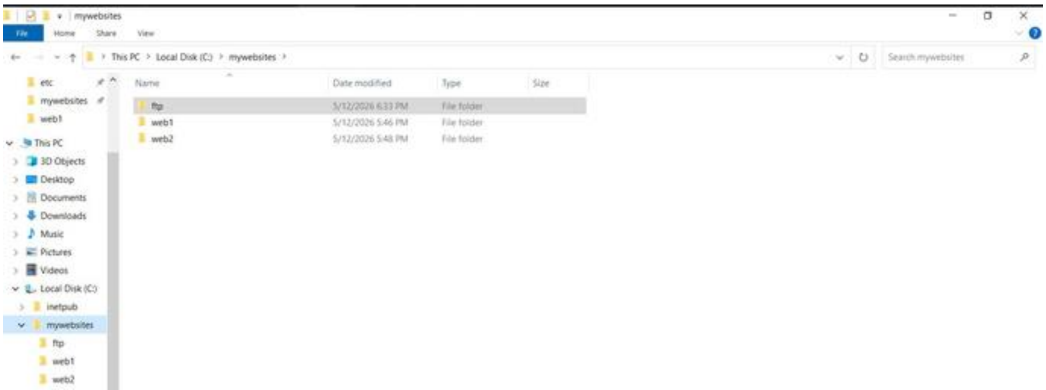


Figure: C:\mywebsites directory containing ftp, web1, and web2 subfolders

8.6 FTP Uploaded Folder

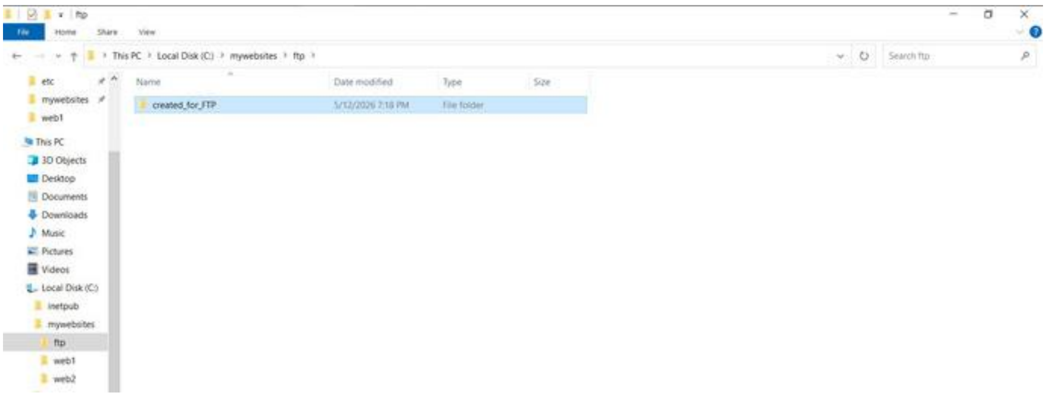


Figure: "created_for_FTP" folder uploaded via FTP client — confirming write access to the FTP site

9. Windows Deployment Services (WDS)

WDS was configured on the Ismaliya server to enable PXE (network) boot and remote OS installation. Four Windows Server install images and one boot image were uploaded. A client PC successfully booted over the network and installed Windows via WDS.

9.1 WDS Boot Image

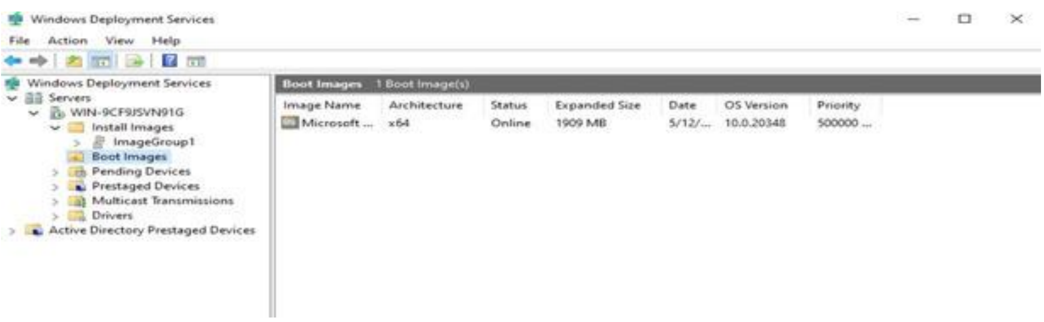


Figure: Boot Images section — Microsoft WinPE x64 (1909 MB) — OS Version 10.0.20348, Status: Online

9.2 WDS Install Images

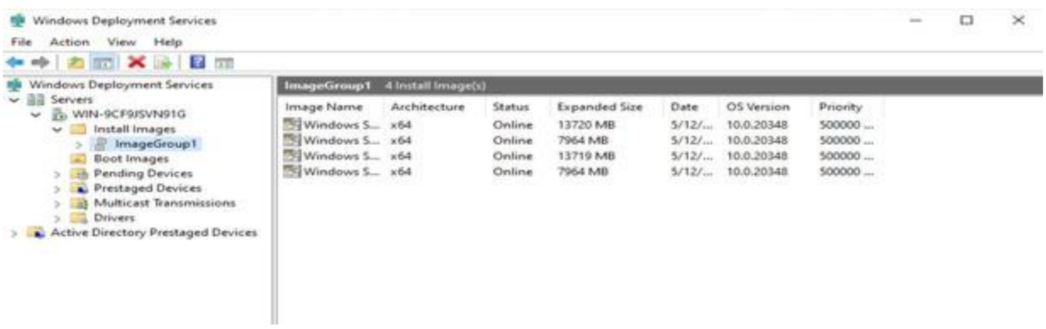


Figure: ImageGroup1 containing 4 Windows Server install images (13720 MB / 7964 MB variants) for deployment

9.3 WDS Setup — Deprecation Notice

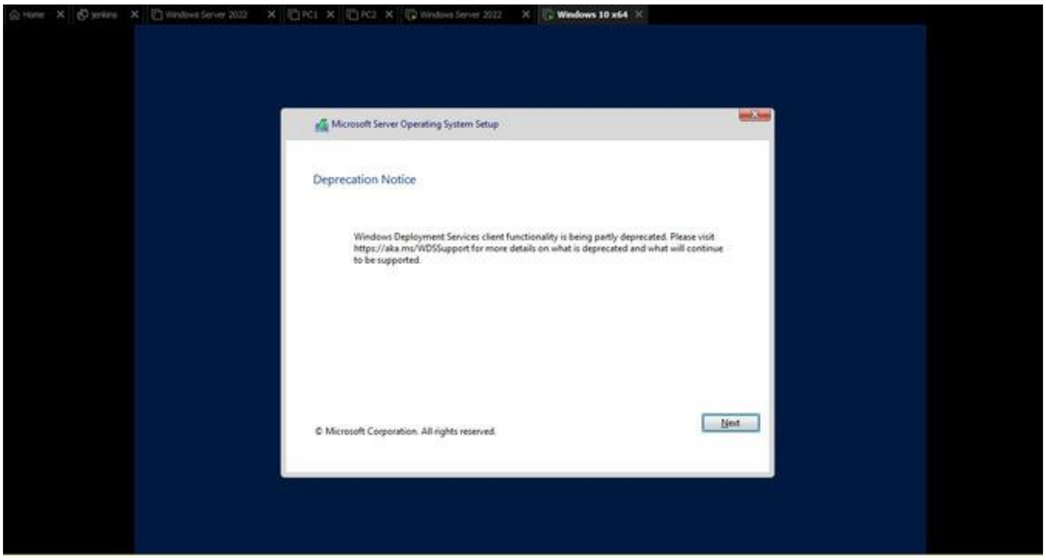


Figure: WDS client setup wizard — Deprecation Notice displayed (WDS client being partly deprecated in newer Windows)

9.4 WDS Setup — Authentication

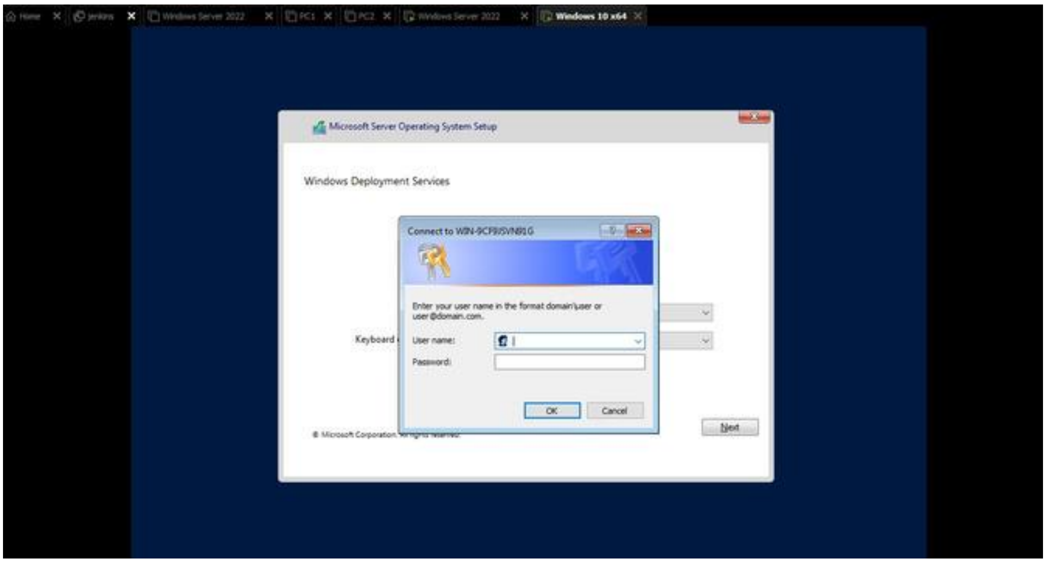


Figure: WDS authentication prompt — connecting to WIN-9CF9JSVN91G with domain credentials to start deployment

9.5 OS Installation via WDS

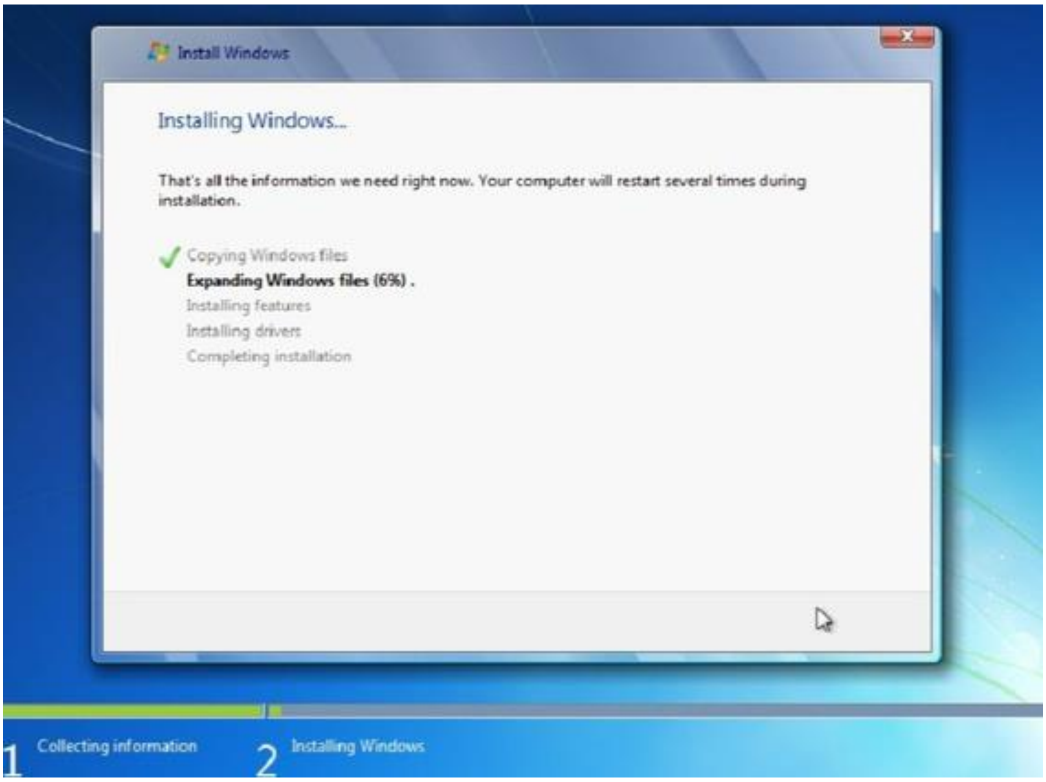


Figure: Windows installation in progress over the network — "Expanding Windows files (6%)" showing WDS deployment working